NEERAJ MAYEKAR

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SUMMARY

MCA student building production-style full-stack applications with React.js, Node.js/Express, and Python (FastAPI), backed by PostgreSQL and MongoDB. Hands-on experience designing RESTful APIs with JWT authentication and role-based access control, and integrating LLM features such as Retrieval-Augmented Generation into web applications. Seeking a Full Stack Development internship where I can contribute to live industry projects and learn from experienced engineers.

TECHNICAL SKILLS

- **Programming Languages:** Python, Java, JavaScript, SQL
- **Generative AI & RAG:** Retrieval-Augmented Generation, Hybrid Search (BM25 + Dense), RRF, Cross-Encoder Reranking, LangChain, Gemini API, Prompt Engineering, ChromaDB, LangSmith
- **Machine Learning & Data:** Scikit-learn, XGBoost, Random Forest, Pandas, NumPy, Matplotlib, Seaborn
- **Backend & Frontend:** FastAPI, React.js, RESTful APIs, JWT
- **Databases:** MongoDB, PostgreSQL
- **Tools:** Docker, Git, GitHub, Linux, AWS EC2

PROJECTS

Enterprise Policy Intelligence RAG Platform | GitHub

LangChain, Gemini, ChromaDB, FastAPI, React.js

- Built an enterprise policy RAG pipeline over 180 pages of compliance documentation, with a layout-aware PDF parser that preserves Markdown structure to ingest multi-column regulatory tables and nested clauses without context loss.
- Architected a hybrid retrieval pipeline combining dense embeddings (BAAI/bge-small-en-v1.5) in ChromaDB with sparse BM25 via Reciprocal Rank Fusion (k = 60), followed by a cross-encoder/ms-marco-MiniLM-L-6-v2 reranker to maximize clause retrieval precision.
- Implemented structured generation with LangChain + Gemini and strict Pydantic schemas, enforcing citation metadata (document IDs, section titles, verbatim quotes) to prevent ungrounded compliance advice.
- Added an LLM-as-a-judge guardrail that scores factual entailment and flags phantom citations before delivery, achieving 96.1% Hit Rate@3, 0.892 MRR, and 99.8% Faithfulness on 51 benchmark queries.

Cost-Aware IT Support Ticket Router | GitHub

XGBoost, LangChain, Sentence-Transformers, FastAPI, React.js

- Built an end-to-end ticket routing platform generating semantic embeddings via Sentence-Transformers and classifying tickets into 5 categories with XGBoost, achieving 0.74 weighted F1 with class-balanced training on severe imbalance (95 to 2,953 samples per class).
- Designed a confidence-threshold decision system that resolves 64% of tickets independently at zero cost with 85% accuracy, routing ambiguous cases to LangChain + Gemini fallback prompts, cutting LLM calls by 64%; monitored token usage with LangSmith.
- Secured REST APIs with JWT and RBAC for users, queue managers, and admins (SQL storage), and built a human-in-the-loop feedback loop so managers can correct misrouted tickets for future retraining.

EDUCATION

MIT WPU, Pune, India — **Jul 2025 – Present**

Master of Computer Application (Science) | **CGPA: 8.79**

Vishwakarma College of Arts, Commerce & Science, Pune, India — **Aug 2022 – Apr 2025**

Bachelor of Computer Science | **CGPA: 9.04**

CERTIFICATIONS

- Networking Basics — *Cisco*
- Agentic AI Foundations Associate — *Oracle*

ACHIEVEMENTS & EXTRACURRICULAR ACTIVITIES

- Winner & Team Captain at Legacy Cup 2024: Captained Forza Football Club to victory in the 2024 Legacy Cup.
- Runner-up & Team Captain at Troika 2024: Led the team to a runner-up finish at BMCC's football tournament.